MTH2003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Differential Equations

Module Leader: Prof. Vadim N. Biktashev

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

USEFUL FORMULAS

These can be used without proofs, unless a proof is explicitly asked for.

Wronskian of two functions $y_1(x)$, $y_2(x)$ is

$$W[y_1, y_2](x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_1'(x)y_2(x).$$

Abel's identity: if $L = \dfrac{\mathrm{d}^2}{\mathrm{d}x^2} + P(x)\dfrac{\mathrm{d}}{\mathrm{d}x} + Q(x)$ and $L[y_1] = 0$, $L[y_2] = 0$, then for some constant C,

$$W[y_1, y_2](x) = C \exp\left(-\int P(x)\,\mathrm{d}x\right).$$

A simple indefinite integral

$$\int (x+1)\,\mathrm{e}^x\,\mathrm{d}x = x\,\mathrm{e}^x + \text{const.}$$

An important boundary-value problem

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y'(L) = 0$$

has solutions $\lambda = (\pi(n+1/2)/L)^2$, $\phi_n(x) = \sin((n+1/2)\pi x/L)$, $n \in \mathbb{Z}_{\geq 0}$.

Another important boundary-value problem

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(L) = 0$$

has solutions $\lambda = (n\pi/L)^2$, $\phi_n(x) = \sin(n\pi x/L)$, $n \in \mathbb{Z}_{>0}$.

SECTION A

1. (a) Find the general solution $y(x)$ to the following ordinary differential equation:

$$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} - 9y = -\frac{8}{9}\,x^{1/3}, \qquad x > 0.$$

Hence find the particular solution such that $y(1) = 1/5$, $y'(1) = 1/3$. (10)

(b) Verify that $y = \sin(x)$ is a solution of the differential equation

$$\tan^2(x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\tan(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \left(2 + \tan^2(x)\right) y = 0.$$

Find the general solution to this equation. (10)

(c) Use a trial function of the form $y = \sum_{n=0}^{\infty} a_n x^n$ to find the solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2x^2y = 0.$$

Write the general solution in the form $y = Ay_1(x) + By_2(x)$, where $y_1(0) = 1$, $y_1'(0) = 0$ and $y_2(0) = 0$, $y_2'(0) = 1$. Write down the first three non-zero terms of the power series for $y_1(x)$ and $y_2(x)$. (10)

(d) Find *all* eigenvalues λ and the corresponding eigenfunctions for the boundary value problem

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \lambda y = 0, \qquad y(0) = 0, \qquad y'(1) = 0.$$

You may use the relevant result given in the formula sheet, but only to check your answer, not to derive it. (10)

(e) Using the results of part (d), or otherwise, write down the general solution of the partial differential equation with boundary conditions:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \qquad u(0,t) = 0,\ u_x(1,t) = 0,$$

where $u_x \equiv \partial u/\partial x$, in the form of an infinite series of terms of the form $X(x)T(t)$. (10)

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SECTION B

2. (a) Prove Abel's identity (see the formula sheet). (10)

   (b) Find two linearly independent solutions of the equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0.$$

   Verify by direct calculation that Abel's identity holds for these two solutions, and identify the corresponding value of the constant C. (8)

   (c) Write Abel's identity for the differential equation

$$(x+1)y'' + xy' - y = 0.$$

   Verify by substitution that $y_1(x) = \mathrm{e}^{-x}$ is a solution of this equation. Use these two results, Abel's identity and the particular solution $y_1 = \mathrm{e}^{-x}$, to find the general solution of this equation. (7)

[**25**]

3. (a) Consider the following ordinary differential equation:

$$x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + (1-x)\frac{\mathrm{d}y}{\mathrm{d}x} + \lambda y = 0. \qquad \text{(Eq3)}$$

Explain why $x = 0$ is a regular singular point of this equation. Consider solutions in the form of a power series, $y = x^m \sum_{n=0}^{\infty} a_n x^n$. Find admissible value(s) of m and show that the coefficients a_n satisfy the recurrence relation

$$a_{n+1} = \frac{n-\lambda}{(n+1)^2} a_n.$$

The solutions found in this way depend only on one arbitrary constant, e.g. a_0, whereas the general solution of any linear second-order differential equation must depend on two arbitrary constants. Explain the apparent paradox. (10)

(b) Write down explicitly the first five terms in the power series solution considered in part (a), assuming $a_0 = 1$ (do not simplify the expressions for the coefficients). Determine the values of the parameter λ at which the power series solution found in part (a) becomes a polynomial (these are known as Laguerre polynomials). Write down all such polynomials up to and including the cubic. (11)

(c) Verify by substitution that the coefficients of the polynomial solution of (Eq3) of degree k can be chosen as

$$a_n = \frac{(-1)^n k!}{(k-n)!(n!)^2}, \qquad n = 0, 1, \ldots, k.$$

(4)

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4. Suppose that $u(x, y)$ satisfies the 2D Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

in the rectangular region $0 < x < \pi$ and $0 < y < \pi$, subject to boundary conditions

$$u(x, \pi) = f(x)$$

and $u \equiv 0$ on the other three sides.

(a) Find the solution to this equation as an infinite sum of functions with coefficients that depend on integrals involving $f(x)$. You may use the relevant result given in the formula sheet. (12)

(b) Hence find the solution in the case that $f(x) = 1$ for $x \in (\pi/3, 2\pi/3)$ and $f(x) = 0$ for $x \in (0, \pi/3) \cup (2\pi/3, \pi)$. (8)

(c) Find an expression for $u_x(x, y)$, via term-by-term differentiation of the series found in part (b). Estimate how many terms in this series should be taken to estimate the value $u_x(\pi/2, \pi)$ with the error not exceeding 10^{-3}. (5)

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